



a YAGEO company

STEINMETZ Coefficients

26th January, 2020

MSABG

- to know if we have the STEINMETZ coefficients (or any other loss model) to estimate the losses

【Steinmetz ceremony】

$$P_{cv} = C_m \times f^x \times B^y$$

$P_{cv}[\text{W/m}^3], f[\text{Hz}], B[\text{T}]$

【Constant】

frequency=100~300kHz

Material	Temperature (°C)	C _m	x	y
BH1T	25	0.31	1.55	2.35
	50	0.18	1.59	2.36
	120	0.01	1.91	2.51

- The experimental loss density versus induction data for different frequency ranges of our BH1Ts

TYPICAL

Frequency (kHz)	Bm (mT)	Pcv(kW/m3)			
		-30°C	23°C	80°C	120°C
50kHz	10	0.30	0.11	0.04	0.03
	50	13.2	7.1	3.0	2.0
	100	59.5	35.7	17.8	15.7
	200	252.1	169.0	116.7	132.1
	300	584.4	451.4	363.7	382.5
85kHz	10	0.60	0.27	0.11	0.09
	50	25.2	14.8	7.0	5.3
	100	117.7	75.7	41.7	37.9
	200	487.2	355.3	259.5	276.5
	300	1154.0	926.3	760.2	804.5
100kHz	10	0.75	0.34	0.15	0.13
	50	30.9	17.7	7.9	6.6
	100	135.6	89.6	47.1	43.6
	200	602.0	445.2	329.6	354.5
	300	1432.0	1157.0	958.2	1022.0

Confidential

*Thank
you*

